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PROBLEMS ON AGES

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Introduction

Problems on ages are a type of quantitative aptitude question that use algebra to solve for unknown ages, focusing on relationships between people across past, present, and future timelines. Common techniques involve defining present ages as variables (e.g.,x), using ratios (3x :4x), and setting up linear equations to calculate specific ages.

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Concept and Definitions

Present Age : Let the current age be ‘x’years.

Future Age :After n years, the age will be (x+n).

Past Age : n years ago, the age will be (x-n).

Age Relationships :If A is n times as old as B, and B’s age is x, A’s age is nx.

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Common Problem Types

Direct Calculation: Solving for a specific age based on direct relationships (e.g., "Father is four times the age of his daughter").

Time Shift: Solving using data from both the past and future (e.g., "5 years ago..." or "10 years hence...").

Ratio-Based: Using given ratios to represent ages (e.g., if ratio of A and B is 3 : 4 , represent as 3x and 4x).

Example

Rajeev's age after 15 years will be 5 times his age 5 years back. What is the present age of Rajeev?

Solution:

Let present age = x

Age 15 years later = x +15 ; Age 5 years back = x-5

Equation : x + 15 = 5(x-5)

X+15 = 5x-25 → 4x=40→x=10

so., 10 years

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Questions

Ques.1 **Ten years ago, the age of mother was three times the age of her son. After ten years, mother's age will be twice that of her son. Find the ratio of their present ages.**

a) 11 : 7

b) 9 : 5

c) 7 : 4

d) 7 : 3

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Questions

Ques.2 Saransh is 50 years old and Nazma is 40 years old. How long ago was the ratio of their ages 3:2?

a) 20 years

b) 30 years

c) 40 years

d) None of these

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Questions

Ques.3 The ratio of the present ages of John and Jim is 5 : 3. Ten years hence it will be 3 : 2. Find the present age of John.

a) 50 years

b) 37 years

c) 40 years

d) None of these

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Questions

Ques.4 A man said to his son, "I was one-third of your present age when you were born". If the present age of the man is 48 years, find the present age of the son.

a) 40 years

b) 60 years

c) 45 years

d) None of these

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Questions

Ques.5 **The ratio of Sara's age 4 years ago and Vaishali's age after 4 years is 1: 1. Presently, the ratio of their ages is 5: 3. Find the ratio between Sara's age 4 years hence and Vaishali's age 4 years ago.**

a) 3 : 1

b) 1 : 3

c) 4 : 3

d) 3 : 4

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Questions

Ques.6 The sum of ages of 5 children born at the intervals of 3 years each is 50 years. What is the age of the youngest child?

a) 4 years

b) 8 years

c) 10 years

d) None of these

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Questions

Ques.7 **A is two years older than B who is twice as old as C. If the total of the ages of A, B and C be 27, then how old is B?**

a) 7

b) 8

c) 9

d) 10

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Questions

Ques.8 A man is 24 years older than his son. In two years, his age will be twice the age of his son. The present age of his son is:

a) 14 years

b) 18 years

c) 20 years

d) 22 years

Questions

Ques.9 The present ages of three persons in proportions 4 : 7 : 9. Eight years ago, the sum of their ages was 56. Find their present ages (in years).

a) 8,20,28

b) 16,28,36

c) 20,35,45

d) None of these

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Questions

Ques.10 **Ayesha's father was 38 years of age when she was born while her mother was 36 years old when her brother four years younger to her was born. What is the difference between the ages of her parents?**

a) 2 years

b) 4 years

c) 6 years

d) None of these

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Questions

Ques11. Q is as much younger than R as he is older than T. If the sum of the ages of R and T is 50 years, what is definitely the difference between R and Q's age?

a) 1 year

b) 2 years

c) 25 years

d) Data inadequate

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Questions

Ques12. What is Ravi's present age?

I. The present age of Ravi is half of that of his father.

II. After 5 years, the ratio of Ravi's age to that of his father's age will be 6 : 11.

III. Ravi is 5 years younger than his brother.

a) I and II only

b) II and III only

c) All I, II and III

d) I and III only

Questions

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Ques13. What is Tanya's present age?

I. The ratio between the present ages of Tanya and her brother Rahul is 3 : 4 respectively.

II. After 5 years, the ratio of Tany's age and Rahul's age will be 4 : 5.

III. Rahul is 5 years older than his Thanya.

a) I and II only

b) II and III only

c) All I, II and III

d) I and III only

e) Any two of three

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Thank You